

Rubik Representation Trilogy

One-Page Project Overview

This project studies the 228-dimensional cubie representation of the Rubik’s Cube group as a finite, reproducible test case for spectral decomposition, generator-resolved transport, and Lie-closure phenomena in finite-group representations.

The Rubik’s Cube group provides a finite but highly noncommutative representation with rich internal block structure, making it a useful laboratory for studying how spectra, sector decompositions, and accessibility relations interact in an explicit finite system.

The central object is the averaging operator

$$A = \frac{1}{|S|} \sum_{s \in S} \rho(s),$$

where S is the standard 18 face-turn generator set and ρ is the unitary cubie representation. Although the generators do not commute, the spectrum of A collapses to six rational layers:

$$\text{Spec}(A) = \left\{ 1, \frac{8}{9}, \frac{7}{9}, \frac{2}{3}, \frac{5}{9}, \frac{1}{3} \right\}.$$

The trilogy asks what structure lies behind this collapse, how the resulting spectral sectors exchange amplitude under generators, and why discrete composition can create accessibility channels that are absent from the associated Lie-generated closure.

Papers

Paper I: Spectral Sector Decomposition in the Rubik’s Cube Representation. This paper explains the rational spectral collapse of the averaging operator. The mechanism is not generator commutativity, but partition integrality: face-symmetric generator sums force integer trace contributions through block-level spectral structure and cyclotomic cancellation. The paper identifies six spectral layers, their block supports, the missing $k = 5$ layer, and the almost multiplicity-free isotypic decomposition.

Paper II: Noncommutative Transport Topology in the Rubik’s Cube Representation. This paper studies the transport tensor

$$K_{\alpha\beta} = \max_{g \in S} \|P_{\alpha} \rho(g) P_{\beta}\|$$

between nine primitive sectors. In the canonical computation, the observed transport graph is governed by noncommutative block support: most direct edges arise from shared active noncommutative support, with one separate commutative-permutation channel. The result is a sparse sector graph with hub-mediated transport rather than arbitrary spectral mixing.

Paper III: Accessibility Structure in the Rubik’s Cube Representation. This paper compares discrete composition with the Lie algebra generated by matrix logarithms $A_g = \log \rho(g)$. The key observation is that discrete composition can create accessibility channels that remain invisible to the Lie-generated closure. The Lie-generated algebra remains block-preserving, while projector-mediated discrete composition can pass through hybrid sectors. In the Rubik representation, the canonical computation identifies five T7 morphisms: cross-block compositional channels with no direct transport and no Lie-generated accessibility at depth 0 or 1, with higher-depth obstruction following from block support.

Computational Supplement

The Computational Supplement freezes the canonical numerical data, definitions, figures, stability checks, and experiment-to-claim map. It records the six spectral layers, nine primitive sectors, transport matrix, noncommutative support table, commutant dimensions, S_3 negative controls, tolerance sweeps, random-seed sweeps, and figure registry.

The public code computes the representation, projectors, sectors, transport tensors, commutants, Lie diagnostics, figures, and validation tables from source. The intent is that every numerical claim in the papers is reproducible by an explicit script or test.

Scope and Status

The project is not a cube-solving method and does not use search heuristics, pruning tables, or neural networks. It is a representation-theoretic and algebraic-combinatorial study of a finite group action.

The strongest proved statements concern the block structure, spectral trace identities, rationality mechanism under the stated hypotheses, and Lie-generated block preservation. Several broader principles, including the role of noncommutative simple components across generator families, are stated as computationally verified or exploratory where appropriate. Negative controls using small S_3 prototypes and the $N = 2$ pocket-cube representation are included to separate structural effects from numerical artifacts.

While all computations are performed on the Rubik representation, the broader goal is to understand which observed structures are representation-specific and which reflect more general phenomena in finite-dimensional noncommutative representations.

Keywords

Finite-group representations; Rubik's Cube group; spectral graph theory; association schemes; Bose-Mesner algebras; commutants; primitive sectors; transport tensors; Lie closure; algebraic combinatorics.